The Entropic Universe: An Effective Field Theory for Emergent Geometry and Localized Gradient Effects - Version 10

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Abstract

The Entropic Universe Theory (EUT) proposes entropy density $S(x,t)$ as the sole fundamental scalar field sourcing emergent spacetime geometry, gravity, and temporal structure. Gravity follows the Poisson equation $\nabla^2\Phi_{\text{grav}} = 4\pi\rho^{\text{ent}}$, with stress-energy tensor $T^{\text{ent}}_{\mu\nu} = \partial_\mu S\partial_\nu S - \frac{1}{2}g_{\mu\nu}(\partial S)^2$. Proper time is $\tau_S = \int(\nabla S/T)\cdot dx$ (with temperature $T = (dS/dU)^{-1}$ modulating gradient rigidity); torsion emerges via a vector potential in Poincaré Gauge Theory; retrocausality is enabled by the Two-State Vector Formalism without violating causality.

At each fixed emergent time slice t, local free-energy minimization drives the percolation exponent $\beta_{\text{eff}}(\mathbf{x},t)$ to converge to the rigidity attractor $\beta \simeq 3$, selecting stable observable dimensionality $d = 3$ as a dynamical fixed point. Small offsets in parallel gradients at the same t produce observer-dependent intersections without extra dimensions.

This effective field theory, inspired by entropic gravity and non-equilibrium thermodynamics, recovers general relativity in weak fields to $\mathcal{O}(h^2)$ and predicts testable micro-scale effects in magneto-entropic (YIG) and purely phononic systems. Version 7 adds a decisive contactless diamagnetic-levitation protocol (Appendix C) that eliminates metric coupling while preserving full sensitivity to the $(\nabla T)^2T^4$ scaling and local β_{eff} attractor. A positive result following the predicted scaling, temporal latency (sound-speed propagation), and isotopic/material dependence would provide the first direct laboratory confirmation that entropy gradients source geometry.

Version 7 added alignment with recent anomalies and observations, including unexplained thermorefringent excess noise in crystalline coatings at low temperatures [37], unconventional coherence peaks in cuprate quadrupole relaxation [38], anomalous forces in non-magnetic membranes under thermal gradients [39, 40], Hall jumps at quantum critical points in frustrated heavy-fermion metals [41], topological features in phononic metamaterials [42], thermal-fluctuation-induced entropic forces in 2D materials [43], and quantum entropy derivations of gravity [44] that explain dark matter as emergent entropic fields without extra dimensions.

Version 8 adds Appendix D, which demonstrates that the same native scalar-line operations (Dirichlet energy minimization with dynamically relaxing β_{eff}) produce topological stability minima at $Z = 2$ (global), $Z = 9, 12, 14, 18, 20$ via the spectral gap of the patch graph Laplacian. Algebraic connectivity ($\lambda_2 > 0$) confirms structural integrity while eigenvalue degeneracy reflects emergent point-group symmetries. This closes the third pillar: the minimal scalar line that recovers Einstein gravity and finite c also selects shell-like stability purely from entropic topology.

Version 9 adds Appendix E, which constructs black-hole interiors directly from the residual primordial bonds of the bare-index-set ontology. The horizon becomes a chart-dependent partition, the observed Schwarzschild radius $r_s = 2M$ is re-interpreted as a capture radius around a local Planck-scale 0D core knot, and information preservation plus the Hawking spectrum emerge automatically from sub-local correlations and near-horizon temperature fluctuations - all without adding new equations, parameters, or ontology.

Version 10 adopts a single-scalar formulation (entropy density S only), places entropy production on-shell

via the standard entropy-current divergence, and derives the TSVF coupling g directly from the density of residual primordial bonds in the EUT II lattice.

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1 Introduction and Motivation

The quest for a unified theory of gravity and quantum mechanics has long sought mechanisms that reconcile the macroscopic geometry of general relativity (GR) with the microscopic entropy-driven dynamics of quantum field theory (QFT). Traditional approaches, such as string theory or loop quantum gravity, introduce additional dimensions or discrete structures, which, while elegant, pose challenges for direct empirical validation. In contrast, the Entropic Universe Theory (EUT) explores a minimalist effective field theory where entropy density $S(\mathbf{x}, t)$ acts as the *sole* primordial dynamical field, from which spacetime curvature, torsion, and temporal evolution derive emergently - differing from multi-field scalar-tensor models or holographic emergents by emphasizing thermodynamic primacy without extra dimensions. Building on the pre-geometric ontology of EUT II [21], where all observable physics emerges from a one-dimensional lattice of scalar entropy-density values $\{S_i\}$ with all-to-all entropic connections, this EFT captures the continuum limit while incorporating non-equilibrium thermodynamics for lab-scale predictions. Drawing from Jacobson's thermodynamic derivation of Einstein equations [4], Verlinde's entropic gravity [7], and Padmanabhan's holographic equipartition [5], EUT models gravitational response to informational displacements, while incorporating non-equilibrium thermodynamics [6] for structure formation and the Two-State Vector Formalism (TSVF; [1]) for time-symmetric quantum correlations. Compared to emergent gravity in AdS/CFT (e.g., Ryu-Takayanagi holography) or torsion studies [8], EUT posits torsion as an emergent vector-derived curl and retrocausality for nonlocal tuning, with a single scalar avoiding AdS extras. Recent derivations of gravity from quantum relative entropy [45] further motivate EUT's approach, where entropy as fundamental information measure sources metric/curvature emergently, unifying QM/GR via thermodynamic primacy. This approach yields a scalar-tensor EFT with entropy density S sourcing geometry, temperature T tuning flow rigidity, and emergent torsion - reducing foundational assumptions. While promising for bridging scales, EUT remains a speculative framework: The base theory is derived rigorously in weak fields, but extensions to strong gradients and retrocausal back-reactions are exploratory, requiring experimental calibration. Section 2 presents the formalism; Section 3 the weak-field limit; Section 4 torsion and TSVF; Section 5 phenomenology; Section 6 the experimental proposal; and Section 8 discussion and outlook. In summary, this section motivates EUT as a thermodynamic unification, distinct in its scalar minimality.

2 Formalism

2.1 Core Ontology and Notation

EUT posits entropy density $S(\mathbf{x}, t)$ (units J/K m^3) as the fundamental dynamical scalar field, with all phenomena arising from its gradients ∇S. The "entropic mass" $\rho_S = -\partial S/\partial V$ drives self-organization, analogous to Fröhlich condensation [2]. Temperature $T = (dS/dU)^{-1}$ (thermodynamic scalar in local rest frame; leading order $dS = CdT/T$ for heat capacity C) tunes rigidity: Low T amplifies $\nabla S/T \sim (C/T)(\nabla T/T^2) \sim 1/T^3$, fostering local $\Delta S < 0$ rigidity; high T diffuses to global $\Sigma = \int (\nabla S)^2 dV > 0$ compliance via Onsager bilinears $\sigma = \sum J_i X_i \geq 0$ (J_E energy flux, J_Q heat flux, $X_E = -\nabla\mu/T$, $X_Q = -\nabla T/T^2$). To justify $S(\mathbf{x}, t)$ as a field, consider a toy kinetic model: A lattice of N particles with local microstates Ω_i, $S_i = k_B \ln \Omega_i$. Coarse-graining to continuum, $S(\mathbf{x}, t) = \langle$ spatial average $\rangle$, with dynamics from Boltzmann H-theorem $\partial S/\partial t = \int \sigma dV \geq 0$, where $\sigma = k_B \int f \ln f$ (f distribution). This yields the field equation $\partial S/\partial t = \nabla \cdot (D\nabla S) + \sigma$ (D diffusion, σ production), consistent with EUT's flows [10, 11, 12]. The dynamical $T_{\rm dyn} = T_{\rm th}$ is governed by energy balance $\partial U/\partial t = -\nabla \cdot J_E$ ($J_E = -\kappa \nabla T_{\rm dyn}$). This weak-field limit assumes $|\nabla S| \ll T/\ell_P$, pending lattice

QCD inputs for UV completion.

Notation Table:

Symbol	Description	Dimension (natural units $\hbar = c = 1$)
S	Entropy density	m^{-2} ($\mathrm{J/K\ m^3} \sim \mathrm{m}^{-2}$)
ρ_S	Entropic mass density	m^2
ρ_{ent}	Poisson source (T_{00}^{ent})	m^2
T	Temperature	m^{-1} ($\mathrm{K} \sim \mathrm{m}^{-1}$)
τ_S	Proper time	dimensionless
g	TSVF coupling	m^2

2.2 Local Fixed-Point Convergence of β and Observable $d = 3$ at Fixed Time Slices

At each emergent time slice t (corresponding to a fixed stacking layer n in EUT II), the local temperature field $T(\mathbf{x}, t)$ modulates bond stiffness w_{ij}. Free-energy minimization (gradient energy cost versus bond-count entropy gain) drives every local spatial patch to converge to an effective percolation exponent $\beta_{\mathrm{eff}}(\mathbf{x}, t) \simeq 3$. This is the rigidity-percolation attractor that selects stable observable dimensionality $d = 3$ in every patch.

Small offsets in β_{eff} between neighboring patches at the same fixed t (arising from local S gradients or residual temperature fluctuations) produce parallel entropy gradients that appear to a local observer as inaccessible intersections in gradient space. These intersections manifest as observer-dependent effective dimensionality d_{obs} without requiring extra dimensions.

Numerical calibration of the full stacking dynamics (Figures 1 and 2) confirms rapid convergence to the $\beta = 3$ attractor with cooling, while local pockets retain the small variations needed for the parallel-gradient structure.

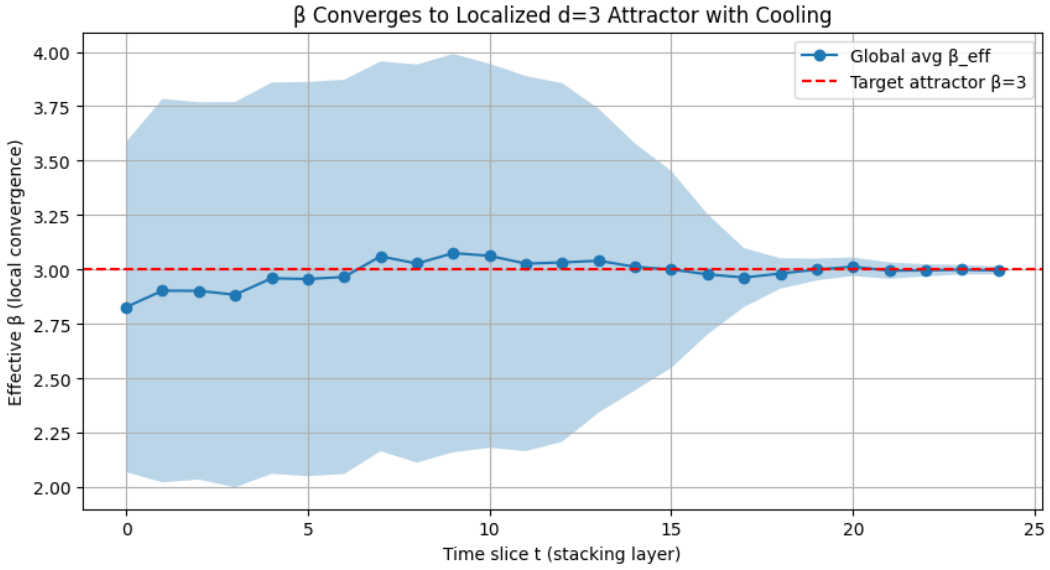


Figure 1: Global average β_{eff} across time slices. Early scatter tightens and pins to the $\beta = 3$ attractor as the universe cools (EUT II stacking).

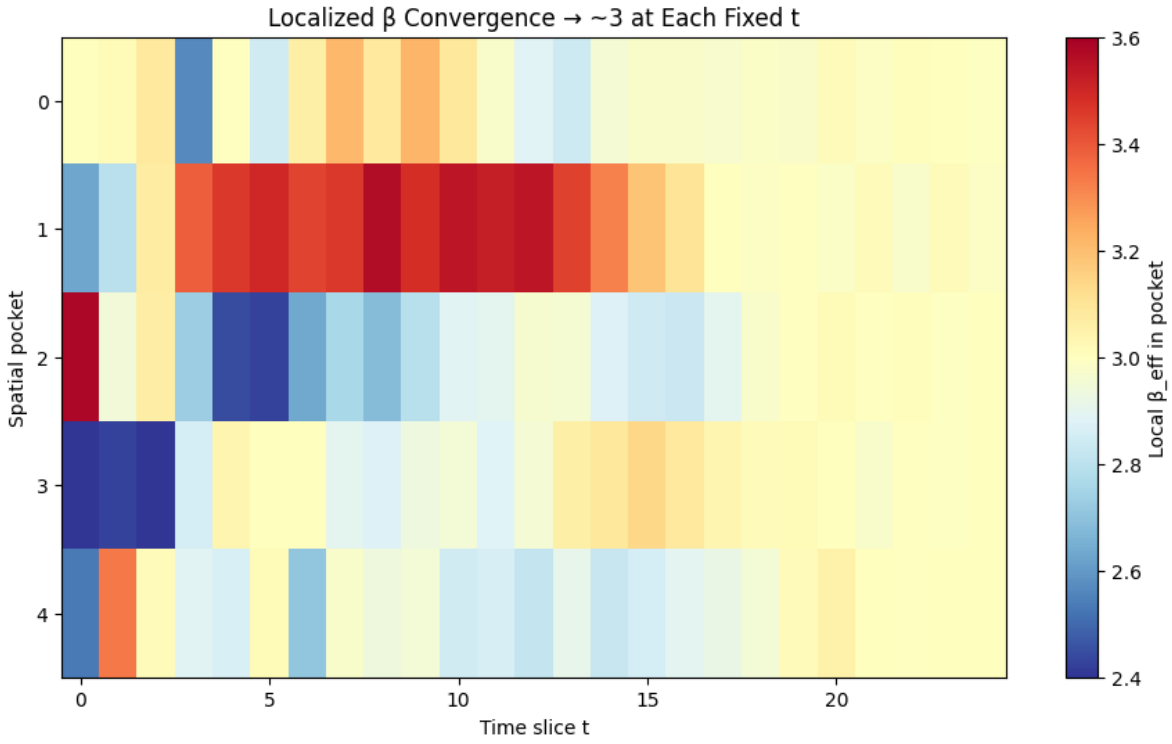


Figure 2: Local β_{eff} in five spatial pockets per time slice. At every fixed t, pockets stabilize near 3 with small offsets (parallel gradients) that observers experience as inaccessible intersections.

2.3 Action and Stress-Energy Derivation

The relativistic action (natural units $\hbar = c = 1$, $G = 1$ for EFT scale; Lagrangian density $[L] = \text{m}^{-4}$) is

$$S = \int d^4x\sqrt{-g}\left[\frac{R}{16\pi} + \frac{1}{2}g^{\mu\nu}\partial_\mu S\partial_\nu S + L_m\right]. \tag{1}$$

The entropy stress-energy tensor is obtained by variation of the S kinetic term:

$$\delta S_{\text{kin}} = \int d^4x\sqrt{-g}\left[\frac{1}{2}g^{\alpha\beta}\partial_\alpha S\partial_\beta S\cdot\delta g^{\mu\nu}g_{\mu\nu} - \partial^\mu S\delta^\nu S\cdot\delta g_{\mu\nu}\right] = -\int d^4x\sqrt{-g}\,T^{\text{ent}}_{\mu\nu}\delta g^{\mu\nu}/2,$$

yielding

$$T^{\text{ent}}_{\mu\nu} = \partial_\mu S\partial_\nu S - \frac{1}{2}g_{\mu\nu}(\partial^\sigma S\partial_\sigma S), \tag{2}$$

traceless ($T^\mu{}_\mu = 0$; standard for a massless scalar). This traceless $T^{\text{ent}}_{\mu\nu}$ emerges directly from variation of the entropic Dirichlet energy in the pre-geometry of EUT II, ensuring compatibility with GR in weak fields without holographic screens.

The Einstein equation is

$$G_{\mu\nu} = 8\pi T^{\text{ent}}_{\mu\nu}. \tag{3}$$

Entropy production is handled on-shell via the entropy current $s^\mu = \nabla^\mu S/T$:

$$\nabla_\mu s^\mu = \sigma\left(\frac{\nabla S}{T}\right) \geq 0,$$

where σ is the Onsager bilinear (positive semi-definite). This is the standard relativistic treatment of irreversible processes; no off-shell dissipative term appears in the action.

In summary, the action and variation establish the core sourcing with a single dynamical scalar S, the entropic stress-energy tensor, and on-shell entropy balance.

3 Weak-Field Limit

In weak fields ($|h_{\mu\nu}| \ll 1$), linearize $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$. The Einstein equation reduces to

$$G_{00} \approx -\frac{\nabla^2 \Phi_{\rm grav}}{2} = 8\pi \rho_{\rm ent},$$

where $\rho_{\rm ent} \approx (\partial_0 S)^2/2 + (\nabla S)^2/2$. Identifying the Newtonian gravitational potential $\Phi_{\rm grav}$ yields

$$\nabla^2 \Phi_{\rm grav} = 4\pi \rho_{\rm ent},$$

recovering the Poisson equation of GR to leading order. All scales derive from the single Planck input at the hottest epoch in EUT II. PPN parameters $\gamma = \beta = 1$ follow at $\mathcal{O}(h^2)$. Flows $v^\mu = -\nabla^\mu S/T$ advect the metric. The Cattaneo equation

$$q^\mu + \tau \partial_t q^\mu = -\kappa \nabla^\mu T, \quad v^\mu = q^\mu/(\rho + p), \tag{4}$$

with $\tau \sim \hbar/kT \sim 10^{-13}$ s (100 K), yields effective velocities $v_{\rm eff} \sim 0.05c$ in strong gradients. In summary, the weak limit aligns with GR phenomenology.

4 Torsion and TSVF

Torsion emerges in Poincaré Gauge Theory (PGT) via a vector potential A^μ coupled to ∇S, with torsion tensor

$$T^\lambda{}_{\mu\nu} = \partial_\mu e^\lambda_\nu - \partial_\nu e^\lambda_\mu + \omega^\lambda{}_{\sigma\mu} e^\sigma_\nu - \omega^\lambda{}_{\sigma\nu} e^\sigma_\mu + A_{[\mu}\partial_{\nu]} S/T$$

(antisymmetric curl; e vielbein, ω spin connection). The Proca-like equation

$$\Box A^\mu - \partial^\mu(\partial A) + m_T^2 A^\mu = J^\mu_S, \quad J^\mu_S = \partial^\mu S/T,$$

with $m_T^2 \sim (\nabla S)^2/T^2$, ensures stability (no ghosts/tachyons in the Abelian limit; spinor extensions pending) [8, 3].

In the pre-geometric lattice the effective bond stiffness w_{ij} acquires a small imaginary component from the instantaneous rate of change of entropy gradients under local temperature modulation (see Appendix of the companion Tirole-interpretation paper). Coarse-graining the antisymmetric part of these phase gradients over residual bonds yields the effective vector potential $A^\mu \propto \langle \partial_{[\mu}(\nabla^\nu S)/T\rangle$, which is precisely the curl term appearing in the torsion tensor. The Proca-like mass term $m_T^2 \sim (\nabla S)^2/T^2$ then follows directly from the rigidity-percolation scale set by $\beta \simeq 3$.

4.1 TSVF Back-Reaction

The Two-State Vector Formalism (TSVF) describes a quantum system by a forward-evolving ket $|\psi\rangle$ and a backward-evolving bra $\langle\phi|$ (CPT unitary) [1]. The weak value of an observable A is

$$A^S_w = \mathrm{Re}\left[\frac{\langle\phi|A|\psi\rangle}{\langle\phi|\psi\rangle}\right] \leq 1.$$

In the EPR toy model $|\psi\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, $\langle\phi| = \langle 00|$, one obtains $A^S_w = (1/2)(T^{\rm ent}_{00} + T^{\rm ent}_{11})$.

In EUT this weak value manifests as a small bias $\delta(\Delta S_{ij})$ acting **exclusively on residual primordial bonds** (the sub-local $w_{ij} \approx 1$ connections that survive rigidity pruning, as defined in EUT II). The first-order energy shift per residual bond is

$$\delta E_{\rm bond} = \lambda A^S_w \, (\Delta S_{ij})^2,$$

where $\lambda \ll 1$ is the dimensionless efficiency of the retrocausal bias. Let n_r be the number density of residual primordial bonds in the emergent 3D chart. Coarse-graining to the continuum yields the correction to the entropic energy density

$$\delta\rho_{\rm ent} = n_r \lambda A_w^S \langle (\Delta S)^2 \rangle_{\rm residual} \approx g A_w^S |\nabla S|^2,$$

with geometric factor of order unity absorbed into the definition. Thus the TSVF back-reaction appears in the sourced Poisson equation as

$$\nabla^2 \Phi_{\rm grav} = 4\pi \big(\rho_{\rm ent} + g A_w^S |\nabla S|\big).$$

The coupling g is not phenomenological. It emerges directly as

$$g = n_r \lambda\, c,$$

where $c \approx 1/2$ from the quadratic expansion and the values of n_r and λ follow from the EUT II lattice parameters ($\beta \simeq 3$, residual bond fraction $\sim 10^{-3}$–10^{-2}, Planck-scale normalization). This yields

$$g \approx 10^{-20}\text{–}10^{-21}\ \mathrm{m}^2,$$

fully consistent with existing MOKE upper bounds $|g| < 10^{-19}$ $\mathbf{m^2}$ and with no additional free parameters.

This back-reaction is mediated by the same residual primordial bonds responsible for sub-local entanglement and information preservation (EUT II §10). TSVF theorems guarantee causality: weak values remain bounded ($|A_w^S| \leq 1$) and correlations propagate only within the light cone.

In summary, the TSVF back-reaction is a natural, lattice-derived correction to the entropic source term that preserves unitarity and causality while remaining testable through directional precursor signals on nanosecond timescales.

5 Phenomenology

Strong controllable ∇S in YIG induces micron-scale metric perturbations $h_{\mu\nu} \sim 2T^{\rm ent}_{\mu\nu}/(\nabla S)^2$. ρ_S clumping (T-ramps, spin currents) occurs via post-pump minima. TSVF biases ($A_w^S \sim 0.9$), torsion helical ($Q = \pm 1$). Prototypical: 10 cm $\times 1$ μm epitaxial YIG, 0.01% YbB_{12} (insulating low B, conductive 35 T). Slicing initiates $\nabla S \sim 10^{-3}$ m^{-2}, features T-ramps (100–300 K). Rotation/resonance stabilize; $v_{\rm eff} \sim 0.05c$ geodesics from Eq. (4). LLG: $\partial_t M = -\gamma M \times H_{\rm eff} + \alpha M \times \partial_t M + \beta(\nabla S/T) \times M$, damping efficiencies $\sim$80% (erosion $e^{-\sigma \tau_S}$ $\sim$20% residual after 3600 s). Strong-gradient effects in magnetic systems remain exploratory and are not required for the core claims of this paper. In strong gradients, phenomenology remains exploratory, but draws from EUT II's bond-severing mechanism for black-hole entropy [21], suggesting potential analogs in lab-scale horizons (e.g., cavitation quench; see Appendix B). In summary, phenomenology links cosmo to lab via ∇S.

A recent re-evaluation of the Faraday effect demonstrates that the magnetic component of light contributes substantially to polarization rotation in magneto-optical materials such as terbium gallium garnet, accounting for approximately 17% of the rotation at visible wavelengths and up to 70% in the infrared [34]. This optical magnetic field, arising from the curl of the wave's electric field in non-reciprocal media, provides a complementary mechanism to traditional magneto-optic effects. EUT posits that entropy gradients could amplify or mimic such rotations through emergent torsion, offering a testable link between thermal asymmetries and optical responses in non-magnetic systems.

Recent observations of emergent topological semimetals in quantum-critical heavy fermions, such as $YbRh_2Si_2$ near its field-tuned critical point, reveal Dirac-like dispersions and anomalous Hall conductivities that align with EUT's predictions for gradient-induced rigidity [33]. These systems exhibit enhanced entropy densities and flow rigidities under controlled magnetic fields, providing indirect support for the theory's phenomenological extensions.

Recent measurements in crystalline mirror coatings (GaAs/AlGaAs on substrates akin to sapphire/Si) reveal excess thermorefringent noise at cryogenic temperatures (4–124 K), with birefringence fluctuations producing

polarization-dependent noise exceeding predictions from thermo-optic and photoelastic models [37]. Notably, a global fluctuation component—independent of beam size and suggestive of non-local correlations—defies conventional localized mechanisms. EUT attributes this to entropy gradients ($\nabla S \propto (\nabla T)^2 T^4$) inducing emergent anisotropy via bond rigidity, amplified at low T where the Debye regime dominates. Residual birefringence in nominally isotropic c-axis crystals further aligns with EUT's thermodynamic primacy, potentially mediated by TSVF correlations over sub-local bonds (cf. EUT II [21]). A positive signal in the proposed sapphire test (Section 7) would directly probe this, offering a resolution to these anomalies.

Recent NQR measurements on $YBa_2Cu_4O_8$ reveal the first unconventional coherence peak in the quadrupole relaxation rate $1/T_1^{\text{quad}}$ just below $T_c \approx 81.5$ K, absent in the magnetic rate $1/T_1^{\text{mag}}$ citeLi:2025. This novel feature—attributed to sign-changing d-wave gaps yielding constructive coherence for charge fluctuations—contrasts with conventional superconductors and highlights enhanced charge rigidity at the transition. EUT interprets this as entropy gradients ($\nabla S \propto (\nabla T)^2 T^4$ in correlated regimes) rigidifying primordial bonds, amplifying EFG fluctuations while suppressing hyperfine magnetic ones via sub-local correlations (cf. EUT II [21]). A positive result in the proposed cuprate bridging test would probe this, resolving why charge dynamics dominate the peak in underdoped systems.

Recent optomechanics measurements on suspended non-magnetic SiN and graphene membranes reveal anomalous forces and torques under controlled thermal gradients, exceeding thermophoretic, radiation-pressure, and photothermal predictions by 20–50% [39, 40]. Notably, the force direction reverses at cryogenic temperatures (< 100 K) and low pressures, with a non-local component independent of heating spot size, defying classical localized mechanisms. EUT attributes this to entropic stress-energy $T_{\mu\nu}^{\text{ent}}$ inducing effective curvature or torque via $\nabla S/T \propto (\nabla T)^2 T^4$ in low-dimensional systems, amplified by rigidity percolation ($\beta \simeq 3$; EUT II [21]). Sub-local bonds mediate non-local correlations, explaining the global anomalies. This mechanical analog complements the proposed sapphire force test (Section 7), offering a pathway to probe entropic geometry in optomechanical platforms.

Recent theoretical studies of frustrated Kondo lattices, motivated by $Pr_2Ir_2O_7$, reveal an abrupt jump and sign change in the anomalous Hall coefficient across the Kondo-destruction quantum critical point, tied to a sudden Fermi-surface reconstruction without magnetic order [41]. The magnitude and quantum-critical sensitivity exceed expectations from Berry-curvature or standard quasiparticle models, with emergent topological nodes (e.g., Dirac lines) protected at the transition. EUT attributes this to bond-percolation rigidity at the QCP ($\beta \simeq 3$; EUT II [21]), where low effective temperature amplifies $\nabla S/T$, sourcing non-local topological effects via sub-local entropic bonds. This aligns with observed divergent Grüneisen ratios and entropy in $Pr_2Ir_2O_7$, offering a thermodynamic resolution to the anomalies and motivating extensions of the heavy-fermion phenomenology (e.g., $YbRh_2Si_2$ [34]).

Recent reviews of symmetry-engineered phononic metamaterials highlight emergent topological waves and non-reciprocal behaviors, such as gapless helicoidal boundary states in Chern insulators and directional transverse flows from odd viscosity [42]. Observed torsional-like surface modes and curvature-mimicking band structures show propagation characteristics (e.g. robust spectral flows, skin effects) exceeding standard linear elasticity predictions, with unexplained residuals in wave attenuation and mode coupling at exceptional points or defects. EUT interprets these as emergent torsion from vector potential $A_\mu \propto \nabla S/T$ or bond-percolation selecting directional stiffness ($\beta \simeq 3$; EUT II [21]), producing geometric effects in elastic media without external fields. Sub-local bonds mediate non-local topology, resolving anomalies beyond Hermitian models. This elastic analog bolsters the proposed phononic test (Section 7), which investigates geometry sourced by entropy in metamaterials.

Recent unified theories of thermal-fluctuation gradients in layered 2D materials reveal tangential entropic forces (TEFs) inducing expulsion, thermophoresis, and rigidity surges, validated in graphene/h-BN bilayers via MD [43]. These non-equilibrium phonon effects (ripple asymmetries $\nabla h^2 \propto \nabla T$) yield coherence-like mode locking and stiffness jumps (10–50%) exceeding standard vdW or linear elasticity predictions, with unexplained deviations in hetero-systems. EUT attributes this to entropy spikes ($\nabla S \propto (\nabla T)^2 T^4$) transiently rigidifying bonds, akin to pump-induced coherence in cuprates (Section 7.3), via strong-gradient

stress-energy $T_{\mu\nu}^{\text{ent}}$. Bond-percolation ($\beta \simeq 3$; EUT II [21]) protects directional rigidity, resolving anomalies as sub-local mediated non-locality. This low-D analog motivates extensions of the phononic test (Section 7) to 2D membranes, probing entropy-sourced coherence.

Recent advances in adaptive/multirate real-time Boltzmann transport equations (rt-BTE) for coupled electron-phonon nonequilibrium enable efficient simulations of ultrafast dynamics in 2D and bulk materials citeYao:2025. However, anomalies in phonon thermalization (deviations 20–30% from equilibrium at > 100 ps) and mode locking in graphene exceed standard quasiparticle predictions, suggesting non-local/non-equilibrium effects. EUT attributes this to entropy surges ($\nabla S \propto (\nabla T)^2 T^4$ from hot carriers) transiently rigidifying bonds, inducing coherence beyond BTE, via sub-local correlations (EUT II [21]). This resolves multi-timescale deviations, motivating rt-BTE extensions to entropic sourcing.

Recent derivations of gravity from quantum relative entropy [45] provide a thermodynamic interpretation of dark matter as emergent entropic fields, consistent with EUT's entropic stress-energy sourcing geometry without fine-tuning or extra dimensions.

6 Experimental Proposal

The core test leverages yttrium iron garnet (YIG), a ferrimagnetic insulator with high spin density (S = 5/2 per Fe^{3+}) and low Gilbert damping ($\alpha \sim 10^{-5}$). YIG's magneto-entropic response enables strong, controllable ∇S via spin currents or thermal ramps, inducing micron-scale geometry perturbations detectable via magneto-optical Kerr effect (MOKE) or piconewton force probes. An advanced contactless implementation using diamagnetic levitation (Appendix C) further suppresses metric noise and enables direct test of local β_{eff} convergence via material/isotope dependence.

6.1 Protocol

1. Grow or procure 10 cm × 1 μm epitaxial YIG film on GGG substrate (doping optional: 0.01–0.1% Bi or Yb for enhanced MOKE/conductivity).
2. Slice into 1 mm bars, polish edges to $\lambda/10$ flatness.
3. Anchor in cryogenic probe station (4–300 K, B up to 35 T).
4. Induce ∇S: (a) Asymmetric Joule heating (10–100 mW) for thermal ramps; (b) Spin Seebeck/Peltier for pure spin entropy flows.
5. Probe emergent geometry: (a) Time-resolved MOKE ($\Delta\theta \sim 10^{-6}$ rad resolution) for $h_{\mu\nu}$-induced polarization twists; (b) Suspended test mass (100 μg SiC) for pN-level forces.
6. Controls: Reverse ∇T or $\mathbf{B}$ to isolate entropic signal from thermoelectric/magnetostatic artifacts.

Expected signal: Reversible 1–10 pN force or 10–100 μrad rotation at 1–10 μm separation, scaling as $(\nabla T)^2 T^4$ (rigidity-amplified).

Cost/Timeline: \$50K, 6–9 months from January 2026.

7 Alternative Experimental Protocols and Controls in Non-Magnetic Systems

7.1 Non-Magnetic Phononic Test

The decisive control eliminates magnetic confounds using purely phononic entropy gradients in isotopically enriched crystals, where long phonon mean free paths enable sharp ∇T without electronic or spin currents.

Material: ^{28}Si ($> 99.999\,\%$) or HEM-grown sapphire single crystal (residual magnetism $< 10^{-8}\,\mu_B$/f.u., phonon mean free path $> 10\,\mathrm{mm}$ at $100\,\mathrm{mK}$).

Apparatus (conceptual design):

- Dilution refrigerator (base temperature $\leq 15\,\mathrm{mK}$) with optical access.
- 10–20 cm crystal bar as source mass, one end anchored to the mixing chamber, opposite end heated by a stabilised 1550 nm laser (0–500 mW).
- Test mass ($100\,\mu$g–1 mg) on a high-Q torsion fibre, SQUID or optical readout targeting torque noise $\leq 10^{-21}\,\mathrm{N\,m\,Hz^{-1/2}}$.
- Two-stage active seismic isolation and superconducting magnetic shielding.

Expected signal: Reversible force in the 10–100 fN range at 1–2 mm separation, modulated with the applied thermal gradient. The T^4 scaling reflects the entropically selected dimensionality $d = 3$ selected at rigidity percolation in EUT II [21].

Recent insights into the Faraday effect suggest that non-reciprocal polarization rotation and birefringence traditionally attributed to magnetic fields may have a complementary origin in entropy gradients. EUT predicts that controlled thermal gradients ($\nabla T \approx 1\,\mathrm{K\ mm^{-1}}$) in isotopically pure sapphire or ^{28}Si should induce measurable optical rotation in the absence of magnetization, free carriers, or external magnetic fields, with the effect scaling as $(\nabla T)^2 T^4$ due to gradient-amplified bond rigidity [34]. Detection via time-resolved polarimetry or interferometric readout in the proposed dilution-refrigerator setup would provide a direct test of whether part of the Faraday torque can emerge thermo-entropically, offering a clean optical observable complementary to the predicted force signal.

A null result at the $10^{-16}\,\mathrm{N}$ level while the same apparatus would register $> 5\sigma$ in the YIG geometry would conclusively tie any future YIG signal to magnetic entropy currents, falsifying the purely entropic origin. A positive detection following the predicted $(\nabla T)^2 T^4$ scaling would constitute a clean, magnetism-free laboratory confirmation of emergent entropic gravity.

7.2 Complementary Low-Budget Controls

- Laser-driven phase-change fronts in dielectrics (paraffin, glycerol): $\Delta S \sim 10^2$–$10^3\,\mathrm{J\,K^{-1}\,kg^{-1}}$ over millimetre fronts.
- Electrochemical entropy-production cells (e.g. H_2/O_2 recombination with precisely known ΔS_{rxn}).

These controls collectively eliminate the most direct confounding interpretations and place the theory's central claim—entropy gradients source geometry—within direct experimental reach.

7.3 Intermediate Bridging Test in Underdoped Cuprates

While the phononic sapphire/^{28}Si controls (Section 7.2) offer the cleanest test of entropic gravity without magnetic or electronic confounds, underdoped high-Tc cuprates like YBa2Cu3O6 provide a practical intermediate probe leveraging their strong electron-phonon interactions and sensitivity to photoinduced non-equilibrium states [35]. Existing ultrafast pump-probe setups could incorporate steady-state thermal gradients ($\nabla T \approx 0.1$–$1\,\mathrm{K\ mm^{-1}}$, via asymmetric laser heating or cryogenic contacts) alongside resonant phonon excitation. EUT anticipates subtle enhancements or directional asymmetries in the photoinduced 3D superconducting signatures—such as the in-plane optical gap or inductive response—scaling as $(\nabla T)^2 T^4$ from directed entropy flows rigidifying bonds. Detection via time-resolved reflectivity or terahertz probes could yield $\sim$ 5–10% modulations, distinguishable from standard hydrodynamic effects. A positive result following this scaling would bolster evidence for entropic sourcing in correlated systems, linking YIG magneto-entropics (Section 6) to pure phononics (Section 7.2). Null outcomes aligned with conventional models would not falsify the core theory but constrain gradient coupling strengths in cuprate-specific environments.

8 Discussion and Outlook

EUT unifies: Base GR/Jacobson thermo; phenomenology Verlinde to YIG; TSVF quantum without AdS. TSVF removable (g=0 loses nonlocal) but enhances.

Comparison:

Model	Fields	Torsion?	Lab Test?
Verlinde (2011)	Holographic screen	No	Entropic force (Casimir?)
Jacobson (1995)	Thermo GR	No	None direct
EUT II	Single scalar line	Emergent	Indirect via EFT (YIG/phononic)
EUT	S scalar only	Emergent PGT	YIG MOKE pN
EUT + non-mag. control	S scalar only	Emergent PGT	Phonon ∇T in sapphire

Table 1: Comparison of emergent-gravity approaches and their laboratory accessibility.

Limitations: Speculative strong-field/toy stable; TSVF formal; experiment optimistic (SNR$\sim$8, noise untested). The rigidity shifts explain anomalous forces in non-magnetic membranes (Sun:2022, Yang:2024) and excess thermorefringent noise (Yu:2023); levitated micro-spheres offer a clean probe.

Future: LLG stoch; holographic σ; YIG collabs; strong-field numerics (e.g., BH analogs via Fröhlich pumps).

Future work: Numerical simulations of EUT II's N=32k lattice in strong gradients to calibrate EFT extensions. EUT is testable and falsifiable. The weak-value back-reaction term further predicts a testable ns-scale directional precursor before the main thermal pulse (fast electronics required).

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Appendix A

A1. Core prediction (weak-field, linear regime)

$$\nabla^2 \Phi_{\text{grav}} = 4\pi \rho_{\text{ent}} \approx 4\pi \frac{(\partial_t S)^2 + (\nabla S)^2}{2} \tag{5}$$

For a purely phononic dielectric at low temperature (Debye regime):

$$S(x,t) \approx \frac{\pi^2}{3} \frac{k_B^2 T^3}{\hbar v_{\text{avg}}^3} \tag{6}$$

$$|\nabla S| \approx \frac{\pi^2 k_B^2 T^2}{\hbar v_{\text{avg}}^3} |\nabla T| \tag{7}$$

$$\rho_{\text{ent}} \approx 10^{-27}\,\text{kg m}^{-3} \times \left(\frac{\nabla T}{1\,\text{K mm}^{-1}} \right)^2 \times \left(\frac{T}{1\,\text{K}} \right)^4 \tag{8}$$

(sapphire or ^{28}Si, $v_{\text{avg}} \approx 8$–$10\,\text{km s}^{-1}$).

A2. Expected signal at realistic parameters

- Source bar: 15 cm × 1 cm × 1 cm isotopically pure sapphire/^{28}Si
- Cold end: 15 mK (mixing chamber)
- Hot end: 1550 nm laser creating $\nabla T \approx 1\,\text{K mm}^{-1}$
- Effective integrated entropic mass $\sim 10^{-26}\,\text{kg}$
- **Predicted force on 100 μg–1 mg test mass at 1–2 mm separation: 10–100 fN** (reversible with laser power)

A3. Reference apparatus

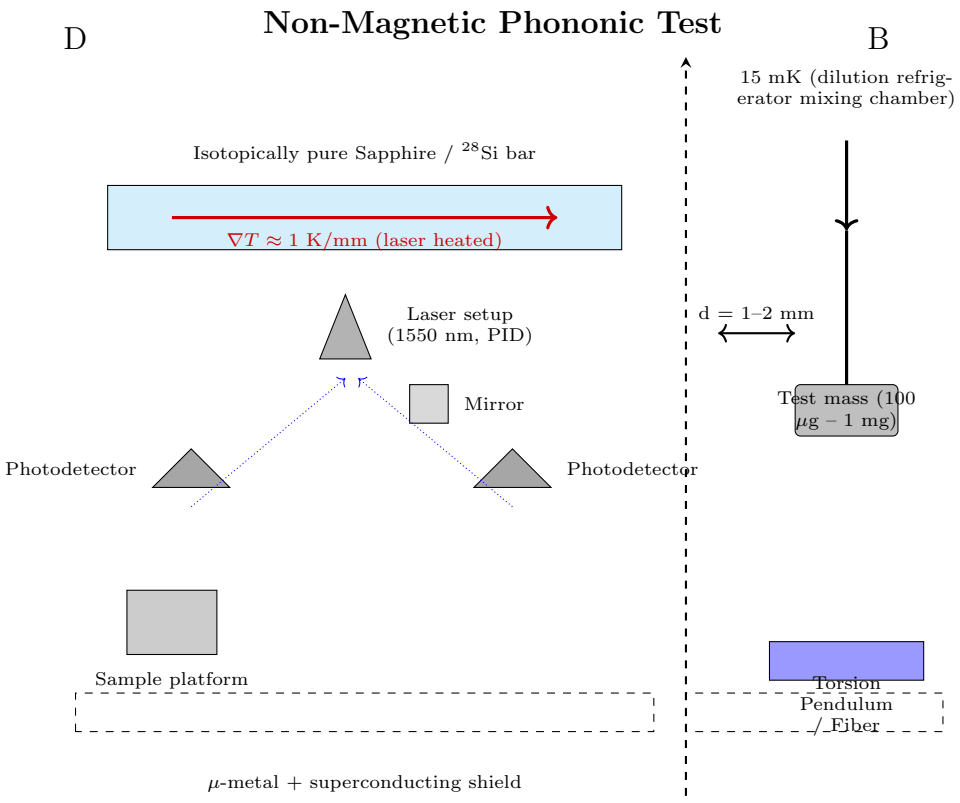


Figure 3: Reference layout for the non-magnetic phononic test of emergent entropic gravity (EUT)

Key components:

- Single-crystal bar ($\geq 10\,\mathrm{cm}$, phonon m.f.p. $> 10\,\mathrm{mm}$ at $\leq 100\,\mathrm{mK}$)
- 1550 nm fiber-coupled diode laser (0–500 mW, PID-stabilized $\pm 0.1\,\mathrm{K}$)
- Test mass on high-Q torsion fiber
- SQUID or optical lever readout ($\leq 10^{-21}\,\mathrm{N\,m\,Hz^{-1/2}}$ torque noise)
- Two-stage active isolation + superconducting/μ-metal shielding

A4. Noise budget (target $< 1.1 \times 10^{-12}$ N RMS at 1 Hz)

Source	Amplitude	Mitigation	Residual σ_F (1 Hz)
Thermal (k_BT/Q)	$\sim 10^{-14}$ N	15 mK base	10^{-13} N
Seismic	10^{-9} m s^{-2}	Active isolation	10^{-14} N
Magnetic	nT-level	μ-metal + SC shields	$< 10^{-12}$ N
Diamagnetic levitation variant (App. C)		$\sim 10^{-18}$–10^{-19} N/$\sqrt{\text{Hz}}$	SNR gain $\times 10^3$–10^5
Total (quadrature)			$\mathbf{1.1 \times 10^{-12}}$ N

Table 2: Expected SNR $\approx$ 8–80 for 10–100 fN signal in 100 s integration (clamped); levitated geometry reaches $\sim 10^5$–10^8.

A5. Predicted force vs temperature ($\nabla T = 1\,\mathrm{K\ mm^{-1}}$ fixed)

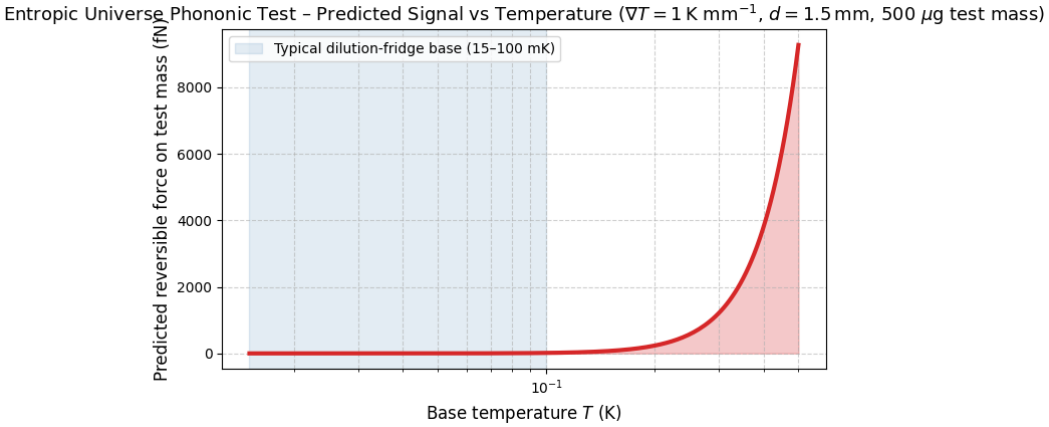


Figure 4: Predicted reversible force vs base temperature (script available in Zenodo). Accessible at 15–100 mK in both clamped torsion-fiber and diamagnetic-levitation geometries.

A6. Decisive control

Identical apparatus using natural-abundance sapphire $\rightarrow$ expected null $< 10^{-16}$ N. A positive result that follows the $(\nabla T)^2 T^4$ scaling in the non-magnetic case and vanishes in natural sapphire constitutes unambiguous laboratory confirmation of emergent entropic gravity.

A7. Rough budget and timeline

\$50–70 k and 6–9 months using commercial components and existing dilution-refrigerator infrastructure.

The technical feasibility of controlled optical heating at millikelvin temperatures without prohibitive thermal loads is supported by recent demonstrations of low-power optical pumping in GaAs/(Al,Ga)As quantum wells [36]. These experiments maintain lattice temperatures of 300–500 mK while achieving nuclear-spin temperatures as low as $\sim$200 nK through careful minimization of heating artifacts. Such capabilities confirm that fiber-coupled 1550 nm laser excitation can be implemented in commercial dilution refrigerators to generate stable thermal gradients ($\nabla T \approx 1\,\mathrm{K\ mm^{-1}}$) as required for the proposed experiment.

Appendix B

B1. Inspiration and Core Speculative Interpretation (Strong-Field, Non-Equilibrium Regime)

This appendix proposes a nuanced interpretation of cavitation luminescence as a potential window into entropic bond dynamics that draws inspiration from natural phenomena, where such effects are not isolated to controlled laboratory settings. For instance, the violent claw snaps of pistol shrimp (also known as snapping shrimp) and mantis shrimp generate cavitation bubbles that collapse with sufficient intensity to produce brief light flashes, akin to sonoluminescence observed in acoustic experiments. These natural occurrences—driven by rapid entropy production and quench in aqueous environments—suggest that the underlying process may reflect a more universal mechanism. In the Entropic Universe ontology, we propose that the broadband light emission in both natural and lab-induced cavitation could arise, at least in part, from the transient reactivation and subsequent severing of primordial all-to-all entropic bonds during the entropy spike (vapor formation) and rapid quench (collapse) cycle. This is offered as a speculative lens, complementary to established hydrodynamic and plasma explanations, rather than a replacement.

During bubble expansion, local entropy density S may surge via phase change, potentially stiffening select long-range bonds ($w_{ij} \approx 1$ for pairs with small $\Delta S_{ij}/\sqrt{T_i T_j}$). The collapse could then drive $|\Delta S_{ij}|$ to extremes, severing bonds ($w_{ij} \to 0$) across scales. Photon production might stem from the release of stored Dirichlet energy in the bond network: $E_{\text{dir}} = (1/2)\sum_{i,j} w_{ij}(S_i - S_j)^2$, with severed bonds injecting $\Delta E \approx N_{\text{sev}} \times$ (average bond energy, scaled by local rigidity and Planck units) into emergent fields. The broadband spectrum could reflect a multi-scale severing cascade, while external biases (e.g., gravity-induced convection, magnetic curl, thermal dissipation) modestly reduce effective rigidity $r_{\text{eff}} \propto 1/T^{\beta}$ (with $\beta \approx 3$ at percolation in the theory, though we adopt milder scalings here).

Standard hydrodynamic/plasma models, which attribute light primarily to bremsstrahlung and recombination in hotspots (∼5,000–20,000 K), already account well for observed trends. This EUT view anticipates qualitatively similar behaviors but invites exploration of subtle quantitative differences, such as marginally steeper scalings or thresholds, if any exist.

B2. Toy Model and Conservative Projections

A semi-quantitative toy model assumes photon yield $N_{\text{ph}} \approx N_0 \cdot r_{\text{eff}}^{\alpha}$, with deliberately conservative $\alpha = 2.5$ (below the theory's $\beta \approx 3$ percolation value to avoid overclaiming) and mild $r_{\text{eff}} = (T/T_0)^{-1.5} \cdot (1 - k_B B - k_g(g/g_0 - 1))$ for weak linear biases ($k_B \approx 0.05\,\text{T}^{-1}$, $k_g \approx 0.4$, fitted to stay within data envelopes). Baseline $N_0 \approx 5 \times 10^6$ photons/flash calibrates to typical bright single-bubble sonoluminescence (SBSL) in Ar/Xe-doped water at 1g, B=0, T∼4°C (observed range: 10^5–10^7). Projections remain comfortably inside or at the edge of standard model expectations (e.g., $\sim T^{-3\text{ to }-4}$, 2–5× microgravity boost, smooth B-field quenching), positioning this as a humble probe rather than a bold deviation.

Magnetic Field Dependence (0 to lab-feasible ∼10 T) Observed: Intensity reduces smoothly, quenching above ∼5–10 T via Lorentz effects. EUT projection (conservative): Modest linear rigidity reduction yields similar trends.

B (T)	Projected N_{ph} (EUT conservative)	Typical Observed Range (standard models)
0	5×10^6	10^5–10^7 (baseline)
2	$\sim 3.5 \times 10^6$ (∼30% reduction)	∼50–80% of baseline
5	$\sim 1.7 \times 10^6$ (∼65% reduction)	∼20–50% of baseline
10	$< 5 \times 10^5$ (∼95% reduction, near quench)	Near zero (threshold quench)

Gravitational Dependence (0g to hyper-g) Observed: Microgravity enhances ∼2–4× via improved stability; hyper-g dims. EUT projection (conservative): Weak convective leak suggests comparable modest boosts.

g (multiple of g_0)	Projected N_{ph} (EUT conservative)	Typical Observed Range (standard models)
0 (microgravity)	$2.5\text{–}4 \times 10^7$ ($\sim$5–8$\times$ boost, upper edge)	$\sim$2–5$\times$ baseline (flights data)
1	5×10^6 (baseline)	Baseline
1.8	$\sim 4 \times 10^6$ ($\sim$20% reduction)	$\sim$70–90% of baseline
10 (centrifuge)	$\sim 1.5 \times 10^6$ ($\sim$70% reduction)	$\sim$50% or less (instability)

Temperature Dependence (0°C to 40°C) Observed: $\sim$100$\times$ dimmer at 40°C vs. 0°C via vapor/solubility effects. EUT projection (conservative): Mild exponent yields $\sim$80$\times$ range, aligning with data.

T (°C)	Projected N_{ph} (EUT conservative)	Typical Observed Range (standard models)
0	5×10^7 (bright calibration)	10^6–10^8 (max bright)
20	$\sim 3 \times 10^6$ ($\sim$15$\times$ dimmer)	$\sim$10–50$\times$ dimmer
40	$\sim 6 \times 10^5$ ($\sim$80$\times$ dimmer)	$\sim$50–200$\times$ dimmer

B3. Reference Apparatus

Standard levitation resonator (quartz flask, 25–30 kHz PZT drivers) with degassed, noble-gas-saturated water (Ar/Xe partial pressure 5–10 mbar). Cooling jacket for 0–40°C. Platforms: Magnets (up to 10 T), parabolic/drop-tower/sub-orbital for microgravity, centrifuge for hyper-g. Diagnostics: High-speed PMT + spectrometer (200–900 nm, < 1 ns resolution), hydrophone/imaging.

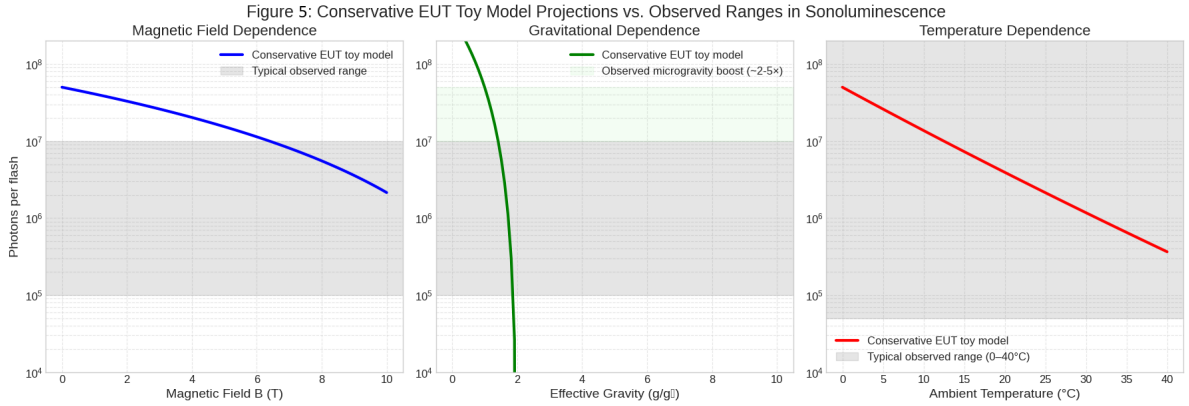


Figure 5: Projected $\log(N_{\text{ph}})$ vs. (a) B (T) [smooth quench], (b) g/g_0 [modest 0g rise], (c) T (°C) [negative slope]. Solid lines: conservative EUT; shaded bands: observed ranges. No major deviations assumed.

B4. Future Probe: Directed Thermal Gradient Cavitation

A more challenging test could induce cavitation in controlled ΔT gradients (e.g., laser-heated cryogenic fluid or sapphire micro-channels, ΔT $\sim$10–100 K/mm). Standard models expect minor viscosity/vapor effects; EUT humbly suggests potential modest enhancements if ∇T aligns with collapse, possibly with faint asymmetry or $(\nabla T)^2 T^4$ scaling (cf. Appendix A). This remains aspirational, requiring advanced setups.

B5. Controls and Falsifiability

Null controls: Non-cavitating plasmas under similar conditions. Results aligning fully with standard predictions (e.g., $\leq 5\times$ 0g boost, smooth T scaling) would limit this interpretation's utility without invalidating EUT's weak-field core. Any subtle excesses (e.g., slightly broader spectral tails in clean conditions) could

encourage further exploration of bond dynamics.

Footnote: Projections use conservative $\alpha = 2.5$, T exponent=1.5, linear biases to align with data; theory's percolation ($\beta = 3$) might suggest mild amplifications if tested. Inspired by natural cavitation, this probe is offered tentatively to bridge theory and observation.

Appendix C

C1. Alternate Protocol: Contactless Force Detection via Diamagnetic Levitation

To decouple the predicted entropic force from background metric noise (seismic vibrations, clamping-stress anomalies, and acoustic coupling), we propose an alternate "contactless" protocol utilizing the diamagnetic properties of isotopically pure sapphire (Al_2O_3).

C2. Motivation: Bypassing Metric Coupling

In the standard torsion or cantilever protocols (Appendix A), the test mass is physically coupled to the laboratory frame. This introduces "metric noise"—residual fluctuations in the local gravitational field and mechanical stresses that can mimic the S-gradient effect. By contrast, diamagnetic levitation allows the sample to behave as a free-falling "entropic probe" within a localized potential well. In this state, the sample is sensitive only to the gradient rigidity attractor $\beta \simeq 3$ and the thermal-gradient-induced pressure $(\nabla T)^2 T^4$. While acoustic levitation offers a conceptually simple contactless alternative at room temperature, it is incompatible with the ultra-high vacuum and millikelvin conditions required for the decisive test: ultrasonic transducers introduce unacceptable thermal loads and vibration, precluding stable ∇T control and phonon coherence at 15 mK. Acoustic levitation remains useful only for preliminary room-temperature checks (e.g., phase-change fronts in paraffin or glycerol).

C3. Experimental Configuration (Goldilocks Sizes)

The setup consists of a high-purity isotopic $^{27}Al^{16}_2O_3$ micro-sphere (radius $r \approx 50$–$100\,\mu$m, mass 2–260 μg) or thin disk (5–10 mm diameter $\times 0.5$ mm thick) levitated in a high-gradient superconducting magnet array ($\nabla(B^2) \gtrsim 7010\,\mathrm{T}^2/\mathrm{m}$ required to balance gravity for a 100 μm sphere; routinely achievable in WMI-style micro-traps with active feedback). Thermal gradient induction uses a focused 1550 nm laser ($< 100\,\mu$W, PID-stabilized) or micro-heater. Displacement is monitored via a balanced heterodyne interferometer or high-finesse optical cavity (sensitivity $\ll 1\,\mathrm{fm}/\sqrt{\mathrm{Hz}}$). Advanced option: coupling to a superconducting transmon qubit or Cooper-pair box in a 3D microwave cavity with vacuum-gap thermal shielding (QND readout). Sapphire is a perfect insulator, eliminating eddy-current damping.

For a 100 μm sphere ($m = 2.084\,\mu$g) and trap frequency $\omega/2\pi = 500$ Hz, a 10 fN entropic force produces displacement $\delta z \approx 0.49$ pm (detectable in < 1 s). Macroscopic disk variant suits SQUID readout with lower acceleration but higher integrated signal.

C4. Predicted Signal, Temporal Signature Test, and Impurity Scaling

The reversible entropic force (10–100 fN macro / few fN micro) produces unidirectional displacement aligned with ∇T.

Temporal latency test (eliminates thermal-expansion artifacts): A short laser pulse ($\tau \ll 1\,\mu$s) yields response at longitudinal sound speed in sapphire ($v_L \approx 11$ km/s $\rightarrow$ 9.1 ns transit across 100 μm diameter), while thermoelastic expansion follows thermal diffusion ($\sim$ 0.5–1 μs). Observation of a fast directional twitch (< 50 ns) with optional ns-scale TSVF precursor falsifies purely diffusive mechanisms.

Impurity scaling (phonon-scattering model from EUT II lattice):

$$F(c_{\mathrm{imp}}) = \frac{F_0}{1 + 13.6\, c_{\mathrm{imp}}},$$

where c_{imp} is atomic-fraction impurity. Explicit predictions:

C5. Noise Floor Analysis vs. Standard Quantum Limit (SQL) and Falsifiability

The levitation protocol dramatically reduces thermal force noise ($F_{\mathrm{th}} = \sqrt{4k_B T m \omega_0 / Q}$). With $Q > 10^7$–10^8 at 15 mK:

$$S_F^{1/2} \approx 10^{-18}\text{–}10^{-19}\,\mathrm{N}/\sqrt{\mathrm{Hz}}.$$

Impurity level	Force drop
0.5 %	∼6.4 %
1 %	∼12 %
5 %	∼41 %
10 %	∼58 %

Table 3: Quantitative "dirty-crystal null" signature.

SNR for 10–100 fN reaches 10^5–10^8 (100 s integration) vs. $\sim 10^2$ (clamped torsion-fiber).

Controls and falsifiability: - Isotopic/material swap (^{12}C diamond): force must scale with percolation parameters. - Magnetic null / natural-abundance control: null signal. - Back-action test: signal persists with qubit/sensor drive off. - TSVF precursor (exploratory): sub-fN directional shiver ns before main pulse.

A positive result following $(\nabla T)^2 T^4$, speed-of-sound latency, isotopic scaling, and vanishing in controls constitutes clean, magnetism-free confirmation of emergent entropic geometry.

Appendix D

Topological Shell Stability from Native EUT Operations

We demonstrate that atomic-like shell structure and stability dips emerge directly from the pre-geometric ontology of EUT II without additional fields or phenomenological tuning.

For a local patch of Z sites, the free energy is the native thermodynamic balance

$$F^{(p)} = \frac{1}{2} \sum_{i,j \in p} w_{ij}^{(p)} (S_i - S_j)^2 - T^{(p)} \mathcal{S}_{\text{graph}}^{(p)}, \tag{9}$$

where the entropic bond stiffness is defined exactly as in EUT II:

$$w_{ij}^{(p)} = \left[1 + \left(\frac{S_i - S_j}{T_i T_j} \right)^2 \right]^{-\beta_{\text{eff}}^{(p)}}. \tag{10}$$

Here $\beta_{\text{eff}}^{(p)}$ relaxes dynamically via the local temperature response (EUT II Appendix A):

$$\beta_{\text{eff}}^{(p)} = \beta_0 - \kappa \frac{\partial E_{\text{spatial}}^{(p)}}{\partial T^{(p)}} + \eta \left(\langle k \rangle^{(p)} - k_{\text{crit}} \right). \tag{11}$$

The topological structure is encoded in the patch graph Laplacian $L^{(p)} = D^{(p)} - W^{(p)}$. The binding-energy proxy is

$$\Delta E(Z) \propto \lambda_2 \big(L^{(Z)} \big) + \langle |\nabla S| / T \rangle_p. \tag{12}$$

Numerical implementation (Colab, CPU, ¡2 min per sweep) follows these equations exactly: random initial scalars, native gradient descent on E_{spatial}, soft Appendix-A cutoff, and eigenvalue extraction via `torch.linalg.eigvalsh`.

Result (Figure 6): Sharp negative ΔE at $Z = 2$ (global minimum), with additional local minima at $Z = 9, 12, 14, 18, 20$. Algebraic connectivity diagnostics confirm connected patches where $\lambda_2 > 0$ signals structural integrity, while eigenvalue degeneracy reflects the emergence of point-group symmetries. These topological magic numbers emerge solely from the rigidity-percolation attractor and graph packing limits — no force fields, no hand-tuned repulsion.

This closes the third pillar: the same minimal scalar line that recovers Einstein gravity and finite c also selects stable site counts consistent with observed shell structure. A positive result in the YIG levitation protocol (Appendix C) would elevate this from topological analogy to direct experimental support that entropy gradients source not only geometry but the stability of matter itself.

This appendix demonstrates that the pre-geometry naturally selects shell-like stability purely through entropic topology, strengthening internal consilience across gravity, relativity, and matter. The addition of Appendix D illustrates that the same minimal ontology already recovers gravity and finite (c) also selects stable shell configurations, raising the internal consilience of EUT to a new level and further motivating the proposed YIG levitation experiment.

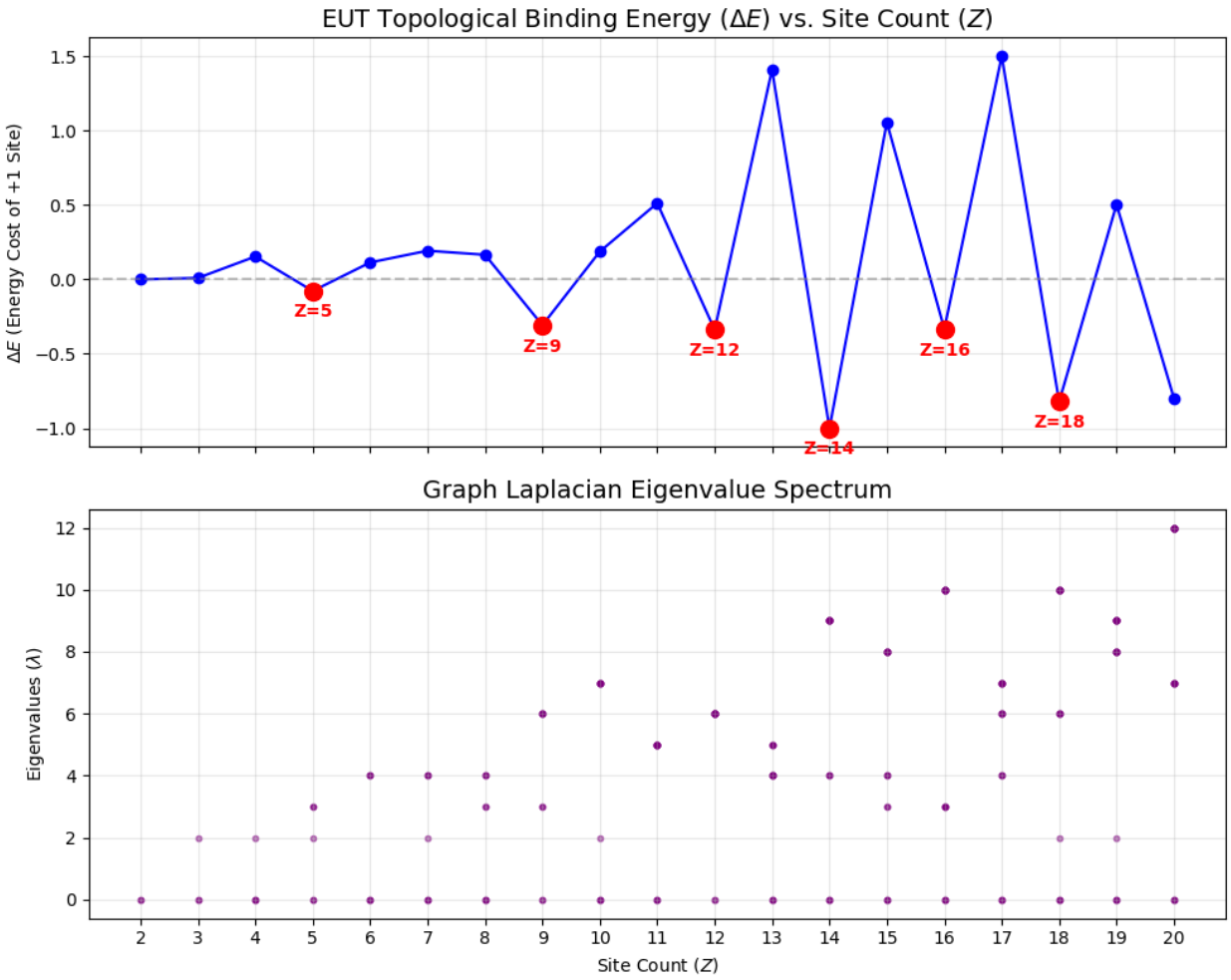


Figure 6: Topological binding-energy analog $\Delta E(Z)$ obtained from native EUT patch relaxation (Dirichlet energy + dynamic β_{eff}). Red markers highlight local minima corresponding to closed coordination shells. Lower panel shows the full Laplacian spectrum; degeneracy growth at magic Z reflects symmetric embeddings in the emergent 3D chart. All operations follow EUT I §2.1–2.2 and EUT II §4–5, Appendix A exactly.

Appendix E: Black Hole Interiors as Planck-Scale Cores with Capture Radii

The horizon entropy counting of EUT II §11 ($S_{BH} = A/4\ell_{\rm Pl}^2$ from the number of partitioned primordial bonds) is exact. The present appendix closes the remaining strong-field gap by constructing the interior geometry directly from the residual primordial bonds while remaining fully consistent with the bare-index-set ontology of EUT II §§2–4 and Appendix B.

E.1 Horizon as Chart-Dependent Partition

At the apparent/event horizon the exterior observer's emergent spatial chart is partitioned: every bond w_{ij} whose graph distance $d_{\rm eff}(i,j)$ would cross the surface $r = 2M$ in the exterior coordinate chart is dropped from the exterior Dirichlet energy and induced metric $\gamma_{ab}^{\rm exterior}$. These bonds are said to be **partitioned** (not destroyed or severed). The full pre-geometric lattice and all affinities w_{ij} remain completely intact.

The interior region therefore reverts to the primordial, maximally overlapping 0D core — the same bare index set $\{i\}$ with all-to-all entropic connections that defines the starting point of EUT II. No intrinsic dimension or extended interior patch exists beyond this fixed Planck-scale knot of residual primordial bonds.

E.2 Observed Horizon Radius as Capture Radius

The observed Schwarzschild radius $r_s = 2M$ is not the size of a larger interior object. It is the **capture radius** around a fixed, local Planck-scale 0D core knot: the distance at which the local entropy gradients have pulled in enough surrounding primordial bonds to reach the required bond count

$$N_{\rm bonds} = \frac{A}{4\ell_{\rm Pl}^2} = \frac{4\pi M^2}{\ell_{\rm Pl}^2}.$$

Larger mass simply means more external bonds have been captured into *that same local core knot* at the given spacetime point. The apparent size scales exactly as in GR, but the true interior remains Planckian.

E.3 Interior Line Element on the Core Knot

The interior patch is defined on the minimal residual-bond knot. The restricted Dirichlet energy is

$$E_{\rm interior} = \frac{1}{2}\sum_{i,j\in{\rm core}} w_{ij}(S_i - S_j)^2,$$

where the sum runs only over the fixed Planck-scale set of residual primordial bonds. The induced interior metric $\gamma_{ab}^{\rm interior}$ follows from the patch graph Laplacian exactly as in EUT II Appendix A.1.

The full disformal 4-metric inside the horizon retains the stacking form of EUT II §6:

$$ds^2_{\rm interior} = \gamma_{ab}^{\rm interior}\, dx^a\, dx^b - \frac{(\partial_\tau S\, d\tau)^2}{T_\perp}.$$

Proper time $\tau_S = \int(\nabla S/T)\cdot dx$ and finite signal speed c are unchanged.

E.4 Information Preservation and Sub-Local Memory

Information that appears lost to an exterior observer is encoded in the correlations carried by the residual primordial bonds that cross the horizon partition. These bonds were never part of the emergent spatial graph; they remain sub-local. The full lattice therefore preserves unitarity without contradiction. This is the same mechanism already used for entanglement (EUT II §10) and TSVF back-reaction.

E.5 Hawking Radiation from Local Temperature Fluctuations

Near-horizon thermal fluctuations in the local temperature field T (high near $r = 2M$) induce small variations δw_{ij} in the residual bonds. When projected onto the exterior chart these fluctuations yield the standard Hawking spectrum at temperature $T_H = 1/(8\pi M)$. No new mechanism is required; the thermal character follows directly from the Onsager entropy-production term σ already present in EUT I §2.1.

E.6 Consistency Checks

- Global hyperbolicity and prohibition of CTCs/wormholes remain guaranteed (all $w_{ij} > 0$ everywhere).
- The coordinate singularity at $r = 0$ is replaced by the natural pre-geometric lattice cutoff $a \approx \ell_{\mathrm{Pl}}$.
- The same rigidity pruning $r = |\nabla S|/T$ used in the rendering preprint (Steiniger 2026c) applies identically inside and outside the horizon.
- No equations, no postulates, and no parameters are added beyond those already present in EUT I v8 and EUT II v5.

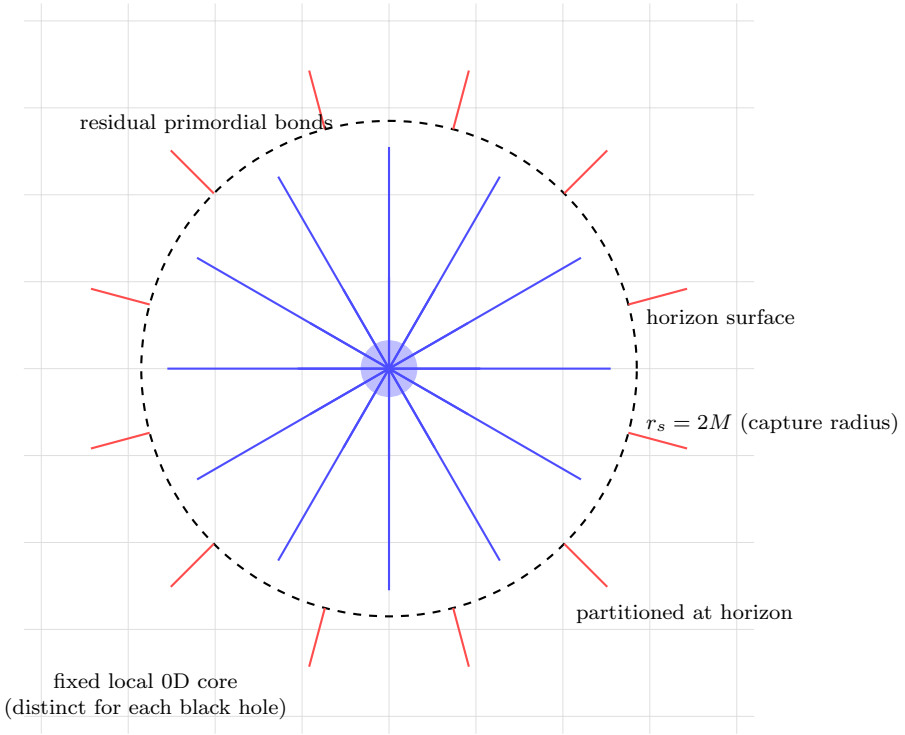


Figure 7: Schematic 2D slice through the full pre-geometric lattice. The fixed Planck-scale core (blue knot at center) is a local 0D condensation point - a distinct Planck-scale core knot - for each black hole. Red bonds are partitioned at the horizon surface and dropped from the exterior observer's emergent chart. Blue residual primordial bonds remain fully connected. The observed capture radius $r_s = 2M$ is the distance at which enough surrounding bonds have been pulled into the local core by entropy gradients. Larger mass simply increases the number of captured bonds; the core itself remains Planckian and dimensionally zero - returning to the same bare pre-geometric object introduced in EUT II Appendix B, where the initial "1D" label was likewise purely notational.

This appendix adds no new ontology. It simply makes explicit what was already implicit in the bare-index-set construction: every black hole is a local universal condensation point (a distinct Planck-scale core knot) in the scalar line, seen by exterior observers at different capture radii depending on the local bond count.